What the Web Shows to AI: Declared versus Enforced Access for Training, Search and User-Triggered Agents

Héctor Pío-Rendón Gutmann
Independent researcher
Germany
h.piorendon.research@proton.me

ABSTRACT

Studies of AI crawler blocking compare an AI user agent's page to a browser's, yet such a request claims an identity it cannot prove. We request the Tranco top 10,000 and a sample of smaller sites with 10 AI agents, neutral controls and spoofed Googlebot and Bingbot, from a residential and a datacenter vantage point. Robots.txt rules and server refusals mostly fall on different sites. Beyond the rate at which sites refuse a spoofed search engine, only training crawlers are refused more often. Where a spoofed user-triggered agent was refused, the real ChatGPT failed on 51.5 % of checks; Claude was mostly stopped by its own fetch tool. HTTP 402 to AI agents carries no price.

CCS CONCEPTS

• **Networks** → **Network measurement**; • **Information systems** → *Web mining*.

KEYWORDS

robots.txt, AI crawlers, web measurement, bot management, pay-per-crawl

1 INTRODUCTION

Website owners are asked to decide whether AI companies may collect their pages. The standard instrument is robots.txt [6]: a site names crawler tokens such as `GPTBot` and the paths they should not fetch. Many studies count these rules [2, 9, 12]. A robots.txt file is a request, though; whom a site actually serves is decided by its server, its content-delivery network (CDN) and its bot management. Recent work therefore also requests pages with AI crawler user agents and compares the answer with a browser's [8, 10, 12].

That comparison mixes three causes. A site may refuse anything that looks like a bot; it may refuse a request that *claims* a known crawler identity but does not come from that crawler's network; or it may refuse AI agents as such. We separate them. Every homepage is requested with a browser user agent, 10 AI agents of five operators in three roles (training, search, user-triggered), spoofed Googlebot and Bingbot, and two neutral controls. A refusal is *AI-specific* only if both controls are served. Spoofed search engines give the baseline: no site means to keep out the real Googlebot, so refusing its name from our address measures identity verification. We find:

- **Declared and enforced blocks mostly fall on different sites.** Of the sites that block an AI agent in robots.txt or AI-specifically on the server, only 9.4–19.1 % (per agent) do both among top sites.
- **Only training crawlers exceed the verification baseline.** On sites that serve both spoofed search engines, training crawlers are still refused on 3.5–7.8 % of top sites, search and user-triggered agents on 1.2–2.5 % and 1.4–2.4 %. Paired tests against spoofed Googlebot reject equality for all training crawlers and for no search or user-triggered agent in the excess direction.
- **Spoofed refusals predict real ones only in part.** Where a site refused the spoofed ChatGPT-User, the real ChatGPT also failed in 17 of 33 checks, against 0 of 14 where it was served. 9 of the 11 refusals the real Claude met came from its own fetch tool, not from the site.
- **HTTP 402 is AI-specific but not a price.** It reaches AI agents on 1.1 % [0.9, 1.4] of top sites; the controls are served on 93.9 % [85.4, 97.6] of them, but only 1 sends a price, which pay-per-crawl gives only to registered, signing crawlers [1].

The study was pre-registered; analyses added later are marked exploratory.

2 RELATED WORK

Declared policy. Longpre et al. [9] audit robots.txt and terms of service behind large AI training corpora and find AI restrictions rising quickly; Fletcher [2] counts AI blocks in robots.txt of leading news sites. Kim et al. [5] observe in server logs that bots follow robots.txt selectively, least for strict rules.

Enforced policy. Liu et al. [8] request the top 10k sites with a headless browser and with two Anthropic user agents, chosen because Anthropic published no IP ranges that sites could verify; they find active blocking common and rarely backed by robots.txt, and examine Cloudflare's AI-bot blocking. Steinacker-Olsztyn et al. [12] compare robots.txt rules and user-agent refusals between reputable news and misinformation sites. An industry report (not peer reviewed) tests GPTBot against a browser on about two thousand sites [10]. These studies compare one AI user agent with a browser. We add neutral and search-engine baselines that separate identity verification from AI policy, three roles per operator, a random sample of smaller sites, two vantage points, and checks with the real agents of the same role.

Real agents and cloaking. Seiden et al. [11] serve canary tokens to scrapers and ask LLMs about them to learn which scraper feeds which model; we use canary pages only to show that an agent fetches live. Serving crawlers a different page is known from search spam; Invernizzi et al. [3] detect it by comparing fetches under several browsing profiles.

3 METHOD

The protocol was frozen before the main run (`PROTOCOL.md`, commit `486ec1f`, archived with the registration) and registered on OSF the same day (https://osf.io/prwng, doi:10.17605/OSF.IO/PRWNG). Analyses not in it are marked *exploratory*; every change is in its deviation log (Section 7).

Population. Tranco list GQ3LK [7], in two strata reported separately: the *head*, ranks 1 to 10,000, and the *tail*, 5,000 domains drawn uniformly at random from the ranks below. A domain is measured only if its robots.txt is reachable, since RFC 9309 reads an unreachable file as "disallow all", and allows our study token. In the head, 2,069 domains failed the first test; 1,372 of them have no DNS entry, 413 refused or timed out the connection, 206 failed TLS and 50 answered with a 5xx status; only the last and some of the timeouts can be refusals of our requests. 2,014 were unreachable from both vantage points. 233 disallowed our token, 126 of them by blocking every crawler name we use; 139 served no HTML. Of 7,559 probed domains, 6,042 were *comparable*: both browser-UA fetches returned the same readable page. In the tail, 3,190 of 3,768 probed domains were comparable.

Agents. Each homepage is requested with 16 user agents: two identical requests with a desktop Chrome user agent (the *browser-UA* baseline), Googlebot and Bingbot, 10 AI agents with the user-agent strings their operators publish, and two controls in the same crawler-style shape whose product tokens are `PageReader/1.0` and `ResearchBot/1.0`. A rule against "bot" in the user agent refuses the second control, so such sites never count as AI-specific. Roles follow the operators' descriptions: training (GPTBot, ClaudeBot, Meta-ExternalAgent, CCBot), search (OAI-SearchBot, Claude-SearchBot, PerplexityBot) and user-triggered (ChatGPT-User, Claude-User, Perplexity-User).

Procedure. We fetch robots.txt with a study user agent that links to an information and opt-out page, then the homepage with the browser user agent. Every other agent then requests its final URL once, in random order, 400 ms apart, without cookies or JavaScript, with no retries. All arms share one HTTP client, and thus one TLS fingerprint.

Classification. A response is *blocked* on a 4xx or 5xx status (except 402 and 429), a bot-management challenge or an empty 202, and only if a later request to the same site still got the page, so that a site that stops answering us is not counted. Content *differs* when the Jaccard similarity of word 3-shingles to the browser-UA page is below 0.80 and at least 0.10 below that of the two browser-UA fetches. Declared policy is a full block of the agent's token in robots.txt under RFC 9309 matching. A block is *AI-specific* when both controls got the browser-UA page. The unit is the comparable domain; brackets give Wilson 95 % intervals.

Baseline test (exploratory). Per AI agent we compare its AI-specific blocks with spoofed Googlebot's on the same sites (exact McNemar test, Holm-adjusted over the 10 agents), and count AI-specific blocks on sites that serve both spoofed search engines, which identity verification cannot explain.

Vantage points and waves. The residential vantage point is a consumer line in Germany (AS3320), the datacenter one a GitHub-hosted runner in Microsoft Azure. Wave 1 measured both strata from both within two days. Wave 2 measured again every domain with a block, a 402 or a content difference in wave 1 (1,876 head and 880 tail domains), one day later.

Real agents. Two rounds, each through the author's consumer accounts, in a fresh chat per check, asking the real ChatGPT, Claude or Perplexity to open a homepage and state its main heading and topic; answers are coded as content, refused/failed or unclear against the browser-UA page. These agents fetch from their operators' networks. The pre-registered round drew head sites with an AI-specific block for GPTBot, ClaudeBot or PerplexityBot. The second (exploratory) round draws, per operator, 40 sites where its spoofed user-triggered agent had an AI-specific block and 15 where it was served; repeats every "content" answer in a fresh chat that forbids browsing, coding it unclear if the model names the heading from memory; and first opens a canary page with a random phrase that exists nowhere else, to show that the agent fetches live. An agent that fails its canary is not tested further. Refusals are split into those by the agent's own fetch tool ("site blocked", robots rules) and those by the site; an answer based on an outdated cached copy is unclear.

Real browser (exploratory). To test the browser-UA baseline, 300 comparable and 300 not comparable head sites, drawn at random, were opened once in Microsoft Edge (Playwright, headless) from the residential line.

4 RESULTS

Unless stated otherwise: head stratum, wave 1, residential vantage point.

4.1 Declared versus enforced

Robots.txt fully blocks at least one AI agent on 13.7 % [12.9, 14.6] of top sites. The server refuses at least one AI agent while serving both controls on 15.0 % [14.1, 16.0]; on 12.1 % [11.3, 12.9] the refused agent is allowed by robots.txt. For GPTBot, 192 sites declare and AI-specifically enforce a block, 449 only declare it and 391 only enforce it (Table 1). Among smaller sites the overlap is 3.7–29.6 %; they declare less (9.6 % [8.6, 10.7]) and enforce more (19.4 % [18.1, 20.8]). Tokens we did not request with are declared at similar rates: Google-Extended on 8.3 % [7.7, 9.0], Bytespider on 11.0 % [10.2, 11.8] of top sites.

4.2 Who is refused

Spoofed Googlebot and Bingbot are almost never blocked in robots.txt (0.1 % and 0.2 %) but refused AI-specifically on 3.9 % [3.5, 4.4] and 4.0 % [3.6, 4.6] of top sites: these sites serve two unknown crawlers yet refuse a known name sent from outside its network. Search and user-triggered AI agents are refused at about this rate (3.1–4.6 % and 3.1–4.6 % per agent), training crawlers more often (5.7–10.2 %).

The overlap per site shows that this is not a coincidence of rates (Figure 1). 45.1–61.6 % of the AI-specific blocks of search agents and 46.8–54.5 % of those of user-triggered agents fall on sites that also refuse a spoofed search engine, against 23.5–39.1 % for training crawlers. On sites that serve both search engines, training crawlers

	robots.txt full block		no full block	
Agent	AI-spec. block	served	AI-spec. block	served
GPTBot	192	449	391	5010
OAI-SearchBot	42	166	197	5637
ChatGPT-User	54	233	224	5531
ClaudeBot	185	379	431	5047
Claude-SearchBot	32	131	153	5726
Claude-User	43	130	146	5723
PerplexityBot	71	270	206	5495
Perplexity-User	47	134	164	5697
Meta-ExternalAgent	71	416	272	5283
CCBot	189	479	323	5051

Table 1: robots.txt full block against AI-specific server block, per AI agent (head, wave 1, residential, comparable sites).

are still refused on 3.5–7.8 %, search and user-triggered agents on only 1.2–2.5 % and 1.4–2.4 %. Paired against spoofed Googlebot, all 4 training crawlers are refused more often ($p < 0.001$, Holm); no search or user-triggered agent is, and 1 search and 1 user-triggered agent are refused significantly *less* often. The same holds among smaller sites (beyond the baseline: training 5.5–12.1 %, search 1.3–2.3 %, user-triggered 1.1–2.3 %) and from the datacenter (training 3.6–7.4 %, search 1.2–2.2 %). Most sites that refuse search and user-triggered agents thus appear to check identities, not to target AI; only training crawlers meet a rule of their own.

4.3 Real agents

In the pre-registered round the real user-triggered agent was refused or failed on 14 of 34 decided checks for ChatGPT, 5 of 34 for Claude and 1 of 11 for Perplexity. Those sites had refused the operator's *crawler*, so the round mixes verification with role. Split by what the spoofed user-triggered agent of the same operator met on the same site, the real ChatGPT was refused on 75.0 % [50.5, 89.8] of sites that refused its spoofed twin and on 12.5 % [3.5, 36.0] of sites that served it; for Claude the shares are 25.0 % [7.1, 59.1] and 12.0 % [4.2, 30.0].

The second round tests this directly. ChatGPT and Claude read their canary pages live; Perplexity (free plan) failed its canary twice, so its round was not run: its answers could not show live access. Where the spoofed ChatGPT-User had been refused, the real ChatGPT failed on 51.5 % [35.2, 67.5] of decided checks; where it had been served, on 0.0 % [0.0, 21.5]. For Claude the shares are 39.3 % [23.6, 57.6] and 8.3 % [1.5, 35.4], but 9 of the 11 refusals came from Claude's own fetch tool, which declined the site before fetching, 6 of them on sites whose robots.txt fully disallows Claude-User; 2 were refusals by the site. Another 4 answers rested on an outdated cached copy. The memory control flagged 0 of 16 ChatGPT and 3 of 20 Claude content answers on these sites. A spoofed refusal thus predicts a real one for ChatGPT on about half of the sites, while the real agent rarely fails where the spoof was served; for Claude, what the real agent meets is largely its operator's own robots.txt compliance, which a spoofed measurement cannot observe. Spoofed shares describe how sites treat traffic that *claims* to be an AI agent.

Stratum	Vantage	Recurring blocks	κ
Head	residential	97.2–99.6 %	0.98–0.99
Head	datacenter	95.7–99.1 %	0.97–0.99
Tail	residential	98.2–100.0 %	0.97–0.99
Tail	datacenter	97.8–100.0 %	0.97–1.00

Table 2: Wave-1 AI-specific blocks recurring in wave 2, range over AI agents. Wave 2 covered flagged domains only, so κ is not a population value.

4.4 Robustness

Browser baseline. On comparable sites the real browser got a readable page on 85.7 % [81.2, 89.2] and the same title as our browser-UA fetch on 79.0 % [74.0, 83.2]; a headless browser is itself refused by some bot management, so it is no gold standard. On not comparable sites our client was refused on 97.3 % [94.8, 98.6], but the real browser got a readable page where ours was refused on only 5.0 % [3.1, 8.1]: few sites drop out because they single out our client; most refuse automated clients of any kind.

Order. Refusals of any kind are as frequent early in the random order (10.6 % [10.3, 11.0]) as late (10.7 % [10.3, 11.0]), so the burst of requests does not itself trigger blocking. 24.7 % [23.7, 25.8] of refusals of AI agents had no later served request and are not counted; since the order is random per site, this lowers all shares alike.

Who refuses (exploratory). Of AI-specific blocks of AI agents, 32.0 % [30.5, 33.6] carry a bot-management signature (challenge page, `cf-mitigated`, CDN block page), 36.7 % [35.1, 38.4] pass a CDN without one and 31.3 % [29.7, 32.8] show neither; CDN headers alone do not show that the CDN refused.

Vantage point and time. From the datacenter, the share with an AI-specific block is 13.9 % [13.0, 14.8]; per-agent agreement between vantage points is $\kappa = 0.67$–0.95 in the head and 0.84–0.99 in the tail. Wave 2, one day later, shows repeatability, not stability over time: almost all AI-specific blocks recur (Table 2).

4.5 Pay-per-crawl

At least one AI agent received HTTP 402 on 1.1 % [0.9, 1.4] of top sites (66 sites) and 0.1 % [0.0, 0.3] of smaller sites. On these top sites the controls were served on 93.9 % [85.4, 97.6] and a spoofed search engine got a 402 on 1.5 % [0.3, 8.1]: the 402 targets AI agents. Only 1 sent the `crawler-price` header. This is expected and says little about a market: pay-per-crawl quotes prices only to crawlers registered with the CDN that sign their requests [1], which a spoofed request cannot do.

4.6 Different content

An AI agent got substantially different content while both controls got the browser-UA page on 2.1 % [1.8, 2.5] of top sites, the same share as for Googlebot or Bingbot (2.1 % [1.7, 2.5]). The planned manual coding was not done (Section 7), so these are upper bounds and we claim no AI-specific content.

5 ETHICS

The study has no human subjects; we followed the principles of the Menlo Report [4]. As an independent researcher the author had no

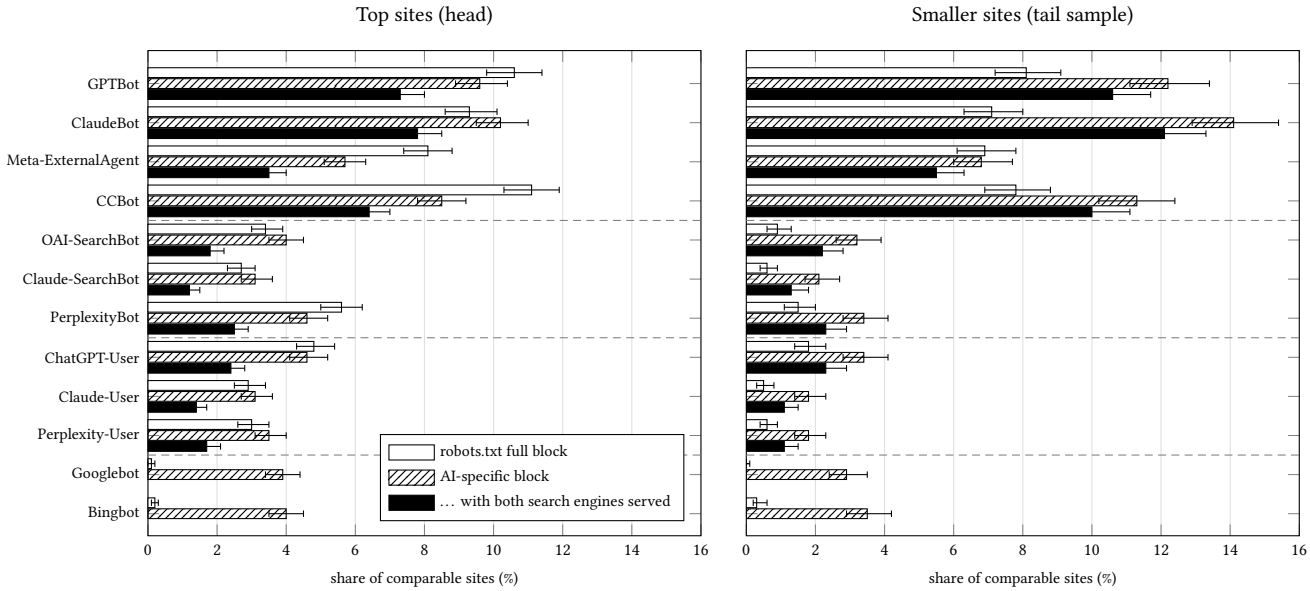


Figure 1: Per agent: robots.txt full block, AI-specific server block, and AI-specific block on sites that serve both spoofed search engines (exploratory), with Wilson 95 % intervals (wave 1, residential). Groups from top: training, search, user-triggered agents, spoofed search engines.

institutional review board; the design was published on OSF before the main run.

Load. One request per agent and homepage, 400 ms apart, homepages only, no retries after an HTTP answer.

robots.txt. We obeyed it for our study token and did not contact sites further whose robots.txt was unreachable. The agent requests, however, reached sites whose robots.txt disallows those agents' tokens. We judged this acceptable because the rules address the operators' crawlers, each such site received one homepage request per token, the request is the minimal observation of enforcement and the established method of prior work [8, 12], and no content beyond page features is kept or released.

Spoofed identities. Our requests carry the names of Google, Microsoft and five AI operators. A site that verifies identities classes them as fake; one that does not may log about a dozen requests from our address as these operators' traffic. We consider this harm small and disclose it here and on the information page. The opt-out link was in the study user agent only, which fetched robots.txt; the agent requests carried the operators' strings. Opt-outs received: 0.

Real agents. The checks ran in the author's own consumer accounts, typed in sequence by a browser assistant under the author's supervision. Automated use of consumer interfaces may conflict with operators' terms; we kept it to the checks reported. Adult sites were skipped.

Release. No site is named. The per-site file replaces each domain with a keyed hash and its rank with a rank band; this stops casual naming, not determined re-identification.

6 LIMITATIONS

Spoofed user agents come from our two addresses only; the real-agent checks are small and through personal accounts. **Homepages only, no JavaScript;** sites that refuse our client drop out as not comparable (Section 4.4). **The robots.txt gate** excludes sites that disallow our token; they declare more AI blocks (204 of 233 block GPTBot), so declared shares are lower bounds. **Content differences** are uncoded. **Time:** two waves one day apart, one residential line in one country. **Exploratory** analyses were chosen after seeing the pre-registered results.

7 DEVIATIONS FROM THE PROTOCOL

(i) Residential runs restarted automatically after native crashes of the runtime; domains in flight were measured again from scratch. (ii) An analysis bug passed the array index as threshold, so all but the first run were analysed without robots.txt parsing; it was found and fixed on the first analysis, before any result was reported. (iii) The real-agent prompt asked for the main heading and topic instead of a verbatim quote, which one agent declined. (iv) The manual coding of content differences was not done. (v) Wave 2 ran one day after wave 1 instead of at least seven days later, and no later wave followed. (vi) The baseline test, the real-browser check, the second real-agent round, the signature attribution and the order analysis were added after an internal pre-review. (vii) In that round Perplexity failed its canary and was not tested, and adult sites were skipped.

8 CONCLUSION

What robots.txt says and what servers do are two policies that rarely coincide on the same site. Measuring the second by spoofing

user agents mixes AI policy with verification of crawler identity. With spoofed search engines as a baseline, only training crawlers meet AI-specific rules; the refusals that search and user-triggered agents meet are mostly identity checks. Real agents bring a third party into view: the operator's own fetch tool, which may decline a site because of its robots.txt before the site can decide. Studies of AI access should report such baselines, test real agents of the same role, and not read a 402 to an unverified request as a price.

ACKNOWLEDGMENTS

The measurement software, the analysis scripts and drafts of this text were written with the help of Claude (Anthropic), a generative AI assistant, under the author's direction. Every reported number is generated from the result files. The author is responsible for the content.

Availability. Code, aggregate results and the hashed per-site file are in the study repository (https://github.com/Nu1111111111/ai-web-measurement). Raw responses are not released (third-party content); the hash key and per-domain files are available to the editor and reviewers on request.

REFERENCES

[1] Will Allen and Simon Newton. 2025. Introducing pay per crawl: Enabling content owners to charge AI crawlers for access. Cloudflare Blog. https://blog.cloudflare.com/introducing-pay-per-crawl/ 1 July 2025.

[2] Richard Fletcher. 2024. *How Many News Websites Block AI Crawlers?* Factsheet. Reuters Institute for the Study of Journalism. https://reutersinstitute.politics.ox.ac.uk/sites/default/files/2024-02/Fletcher_How_Many_News_Websites_Block_AI_Crawlers.pdf

[3] Luca Invernizzi, Kurt Thomas, Alexandros Kapravelos, Oxana Comanescu, Jean-Michel Picod, and Elie Bursztein. 2016. Cloak of Visibility: Detecting When Machines Browse a Different Web. In *IEEE Symposium on Security and Privacy (SP)*. 743–758. https://doi.org/10.1109/SP.2016.50

[4] Erin Kenneally and David Dittrich. 2012. *The Menlo Report: Ethical Principles Guiding Information and Communication Technology Research.* Technical Report. U.S. Department of Homeland Security. https://papers.ssrn.com/abstract=2445102

[5] Taein Kim, Karstan Bock, Claire Luo, Amanda Liswood, Chloe Poroslay, and Emily Wenger. 2025. Scrapers Selectively Respect robots.txt Directives: Evidence from a Large-Scale Empirical Study. In *Proceedings of the ACM Internet Measurement Conference (IMC)*. 541–557. https://doi.org/10.1145/3730567.3764471

[6] Martijn Koster, Gary Illyes, Henner Zeller, and Lizzi Sassman. 2022. *Robots Exclusion Protocol.* RFC 9309. IETF. https://doi.org/10.17487/RFC9309

[7] Victor Le Pochat, Tom Van Goethem, Samaneh Tajalizadehkhoob, Maciej Korczyński, and Wouter Joosen. 2019. Tranco: A Research-Oriented Top Sites Ranking Hardened Against Manipulation. In *Network and Distributed System Security Symposium (NDSS)*. https://doi.org/10.14722/ndss.2019.23386

[8] Enze Liu, Elisa Luo, Shawn Shan, Geoffrey M. Voelker, Ben Y. Zhao, and Stefan Savage. 2025. Somesite I Used to Crawl: Awareness, Agency and Efficacy in Protecting Content Creators from AI Crawlers. In *Proceedings of the ACM Internet Measurement Conference (IMC)*. 78–99. https://doi.org/10.1145/3730567.3732913

[9] Shayne Longpre, Robert Mahari, et al. 2024. Consent in Crisis: The Rapid Decline of the AI Data Commons. In *Advances in Neural Information Processing Systems 37, Datasets and Benchmarks Track.* https://proceedings.neurips.cc/paper_files/paper/2024/hash/c3738949a80306cc48a8ea8ba0560f9d-Abstract.html

[10] Roman Milyushkevich. 2026. The AI Crawler Block Index. HasData blog (industry report, not peer reviewed). https://hasdata.com/blog/ai-crawler-block-index

[11] Steven Seiden, Triss Ren, Caroline Zhang, Taein Kim, Enze Liu, and Emily Wenger. 2026. Identifying AI Web Scrapers Using Canary Tokens. arXiv:2605.13706. https://arxiv.org/abs/2605.13706

[12] Nicolas Steinacker-Olsztyn, Devashish Gosain, and Ha Dao. 2026. Is Misinformation More Open? A Study of robots.txt Gatekeeping on the Web. In *Proceedings of the ACM Web Conference (WWW)*. 7507–7516. https://doi.org/10.1145/3774904.3792625